A sequence is simply an ordered list of numbers.

Ex: 1, 2, 3, 4, 5 .... or 1, 1/2, 1/4, 1/8, ... or 1, -1, 1, -1, 1 .. ..

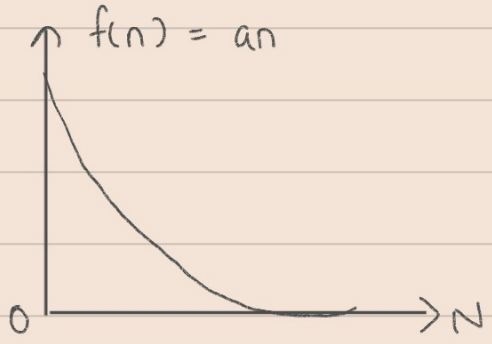
Formal Definition :-

A sequence is a function $f: N \to R$ that is f has domain N and Range R.

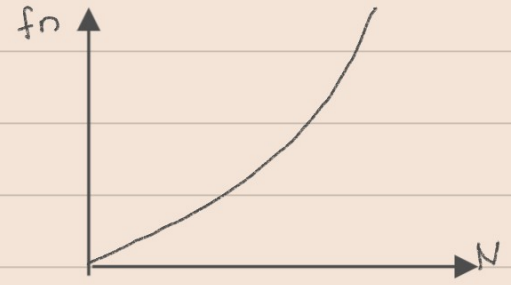
Examples: -

1) $f(n) = \frac{1}{n} = a_n$

As $n \to \infty$ $a_n \to 0$

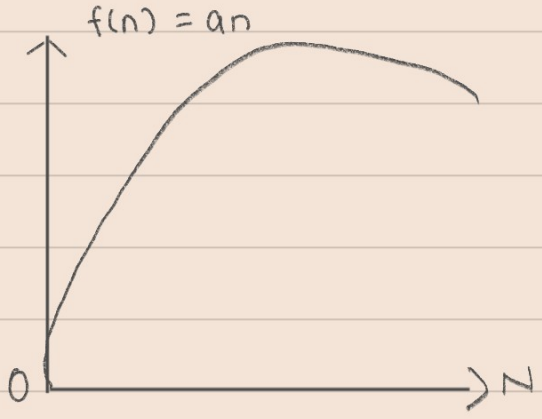


3. $f_n = n^2$



② $f(n) = 1 - \frac{1}{n} = a_n$

As $n \to \infty$, $\frac{1}{n} \to 0$ so $\left(1 - \frac{1}{n}\right) = a_n \to 1$



Neighbourhood

Suppose 'an gets close to a'. What does "close" mean?

We therefore choose a positive number ε and say that we want the terms to be within ε distance of a.

For any $a \in \mathbb{R}$ and $\varepsilon > 0$, the neighbourhood of a is a set of all points whose distance from a is strictly less than ε.

Formally, $B_\varepsilon(a) = \{x \mid |x-a| < \varepsilon\}$

- ε is a positive number representing how much distance from a we allow.
- $|x-a|$ is the distance between x and a.

Therefore, $B_\varepsilon(a)$ = all x whose distance from a is less than ε.

Convergent Sequence.

- We say a sequence converges to a if its terms eventually become as close to a as we want.
- "Eventually" - crucial.

We do not require the first few terms to be close. We only care about what happens in the long run.

If a sequence $\{a_n\}_{n=1}^{\infty}$ converges to a if $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that for every $n \geq N$ $a_n \in B_\varepsilon(a)$.

① Choose a value of $\varepsilon > 0$

② Choose a cutoff N such that for all $n \geq N$ $a_n \in B_\varepsilon(a)$.

The sequence has to pass this test for all values of ε.

Questions

① Prove that:-

ⓐ $a_n = \frac{1}{n}$ converges to 0

Step 1: Start with arbitrary $\varepsilon > 0$

we want, $|a_n - 0| < \varepsilon$

Since, $a_n = \frac{1}{n}$

we need $\left|\frac{1}{n}\right| < \varepsilon$

Since, $n > 0$:

$$\frac{1}{n} < \varepsilon$$

Solve for n

$$\frac{1}{n} < \varepsilon$$

$$n > \frac{1}{\varepsilon}$$

So, $N > \frac{1}{\varepsilon}$

For ex: we can choose

$$N = \left[\frac{1}{\varepsilon}\right] + 1$$

Then for every $n \geq N$:

$$n > \frac{1}{\varepsilon}$$

Therefore: $\frac{1}{n} < \varepsilon$

Hence, $|a_n - 0| < \varepsilon$

Therefore: $\lim_{n \to \infty} \frac{1}{n} = 0$

Meaning of Long Run

For all $n > N$ " captures the idea of the long run.

We are not concerned about what happens at the beginning.

We care about what happens eventually.

For ex: a sequence could begin:

$$1000, -50, 20, 7, 1, \frac{1}{2}, \frac{1}{3}, \ldots$$

and still converge to 0.

For every epsilon, there exists a real number N